Inverse & Composite Functions

Composing functions

1 Given $f(x) = 2x + 3$ and $g(x) = x^2 - 1$, find $(f \circ g)(x)$ and $(g \circ f)(x)$.

2 Given $f(x) = \sqrt{x + 4}$ and $g(x) = x^2 - 4$, find $(f \circ g)(2)$ and state the domain restriction that makes $f \circ g$ defined.

Decomposing a composite function

3 Write $h(x) = (3x - 5)^4$ as a composition $h(x) = f(g(x))$ for two simpler functions f and g.

4 Write $h(x) = \sqrt{x^2 + 1}$ as a composition of two functions.

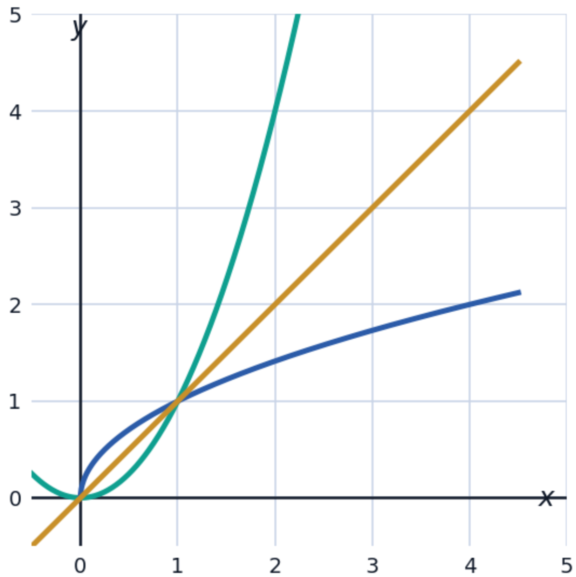
Verifying inverse functions

5 Verify algebraically that $f(x) = \dfrac{x - 2}{3}$ and $g(x) = 3x + 2$ are inverses of each other.

6 Find the inverse of $f(x) = 2x^3 + 1$, and verify that $f(f^{-1}(x)) = x$ for one value, say $x = 17$.

Visualizing inverse functions as reflections

7 The graph shows $f(x) = x^2$ (restricted to $x \geq 0$) and its inverse $f^{-1}(x) = \sqrt{x}$, which are reflections of each other across the line $y = x$.



a State the domain restriction placed on $f(x) = x^2$ needed for it to have an inverse, and explain why.

b Confirm from the graph that f and f^{-1} intersect on the line $y = x$, and find that intersection point algebraically.

Domain and range of composite and inverse functions

8 Given $f(x) = \dfrac{1}{x-3}$ and $g(x) = x + 1$, find the domain of $(f \circ g)(x)$.

9 If f has domain $[1, \infty)$ and range $[0, \infty)$, state the domain and range of f^{-1}.

Applying composite and inverse functions in context

10 A temperature-conversion function $C(F) = \frac{5}{9}(F - 32)$ converts Fahrenheit to Celsius. Find $C^{-1}(F)$ and interpret what it computes.

Using the horizontal line test

11 Determine whether $f(x) = x^3 - 4x$ has an inverse function over its entire domain. Justify using the horizontal line test.

12 Explain why $f(x) = 5x - 2$ passes the horizontal line test for all real x, and find its inverse.

Composite functions with restricted domains

13 Given $f(x) = \frac{1}{x}$ and $g(x) = x - 4$, find $(f \circ g)(x)$ and state its domain.